

XPCyber Challenge Group Analyst Debrief Report

Week #9 XPCyber Challenge - Lengthy Logs: Attack Analysis [NG]

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Term 2

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Executive Summary

This investigation examined reported login failures affecting a WordPress production environment during the XPCyber Challenge: Lengthy Logs – Attack Analysis [NG]. Initial indicators suggested the issue was more than a simple authentication problem, particularly after internal discussion pointed toward the database as the likely source of impact. Based on the observed symptoms and available environment hints, our initial theory was that the underlying WordPress user data or configuration had been altered, causing widespread login issues and possible service degradation.

To validate this theory, we analyzed the database system and reviewed the MySQL log file for suspicious activity. Using targeted searches of SQL queries, we identified unauthorized UPDATE and DELETE operations involving key WordPress tables, including configuration and user-related metadata. We then compared the affected environment against the known-good backup stored on the backup machine to determine the scope of tampering. This comparison confirmed that the database machine was the compromised system, the MySQL service was the affected service, and mysql.log contained the most direct evidence of malicious activity. The analysis also identified four impacted user accounts: playerone, gbates, takasaka, and admin. Based on these findings, we conclude that the incident was a confirmed compromise of the WordPress database environment. The most pragmatic remediation is to restore the database from the verified backup, reset credentials for affected accounts, review database access controls, and conduct additional log review to determine the original point of unauthorized access and prevent recurrence. In addition to the technical recommendations if it is determined that an employee's actions led to unauthorized access, additional training across the board should be implemented. .

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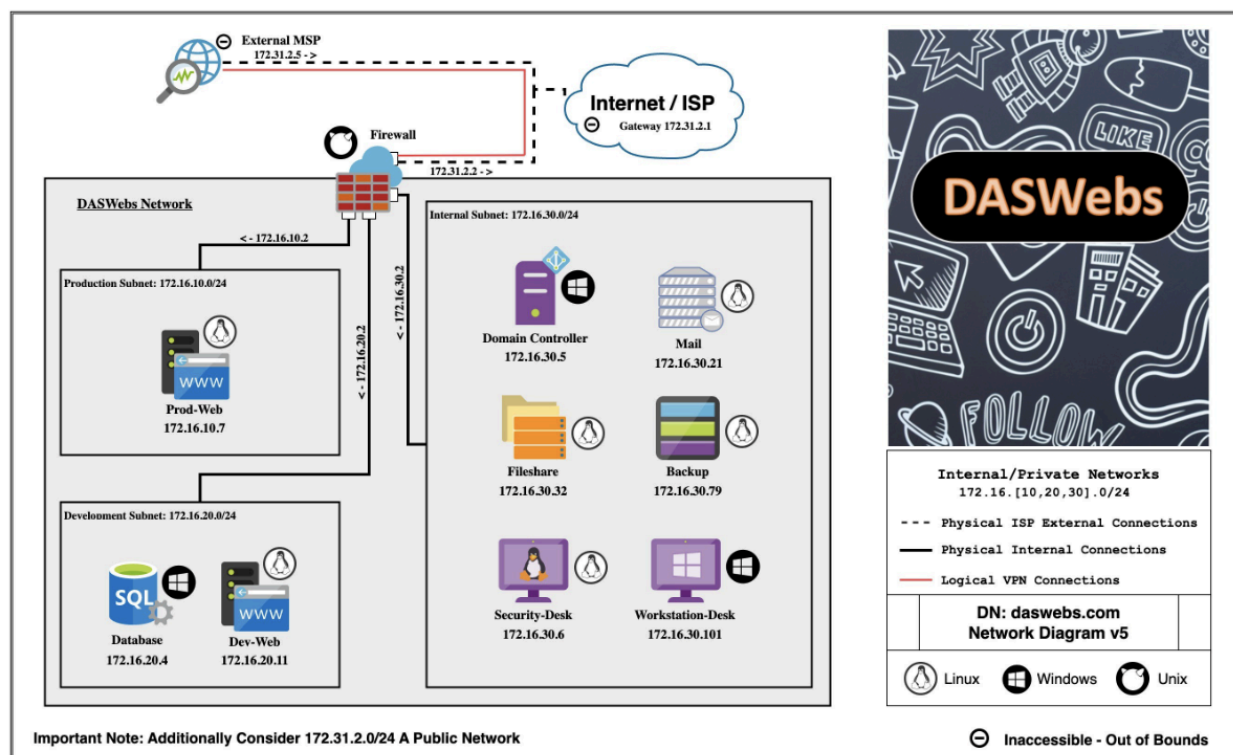
Technical Steps & Analysis

Chat logs

	Rob the Intern I tried to log into the WordPress site we are hosting on Prod-Web and couldn't get in. Not sure what's going on, can someone help? I know I'm typing in my username and password right! I was able to get in yesterday!
	Gilly Bates Did you check your caps lock key? Wouldn't be the first time that's happened.
	Rob the Intern Yes I did! I swear I have it right this time...
	Gary Thatcher Hmm... I actually can't log in to the website either. Something is definitely up. Can we get confirmation by at least one more person?
	Thanh Akasaka Confirmed, no luck for my login on the website either.
	Ione Leventis Of course something like this would happen while I'm out sick. @playerone I'm going to need your assistance here.
	Rob the Intern I don't understand how @playerone can even figure anything out if nobody can get into the website!
	Ione Leventis Take a look at the logs maybe?
	Gilly Bates Yeah and let's not forget our website's user accounts are stored in a MySQL database, on the Database machine. I think it is very likely the database was the victim given the behavior we're seeing with logins.
	Thanh Akasaka Why would anything happen to our database? How would anyone even have access to that box?
	Gary Thatcher Interesting. @playerone start looking into what happened. I'm fairly certain we've been hit by an attack of some sort, but don't worry about figuring out how we were attacked, just figure out how the attack affected us. I'll take care of determining how we were attacked with @ileventis.
	Thanh Akasaka If something is wrong with that specific database can't @playerone just use the mysql backup located in /DatabaseBackup on the Backup machine? I'm certain the backup in there is a known good backup.
	Ione Leventis I mean theoretically yes that might fix the login issues but we need to get to the bottom of what occurred here. @playerone don't worry about restoring the database just figure out what occurred. But the backup will likely be helpful in determining what has changed. The logs for the database service should be in 'C:\mysql_logs' on Database.
	Gilly Bates @playerone, when you find evidence of an attack, we will want you to record some information. We have an Incident Response form you can use to provide us with important details. @gthatcher will give you a bit more information on what we will want to know.
	Gary Thatcher @playerone, we are going to want to know a few different things here. We'll want to know which system was breached; however, we're pretty confident the Database machine was the victim here. Additionally, we'll want to know which service was compromised and the log file that contains the most obvious evidence of the attack. Finally, we will want to know any user accounts that were affected due to the attack. If multiple user accounts were victims, you will need to provide us with the account names on a single report, just separate them with commas.
	XP Cyber Dev Team This challenge requires the use of the challenge webform interface located in the tabbed panel, below the "Virtual Machines" and "Checks" panels. Submitting the challenge webform will affect one, or more, checks. Players will receive a response indicating the correctness of the submitted values after each webform submission.

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Network Map



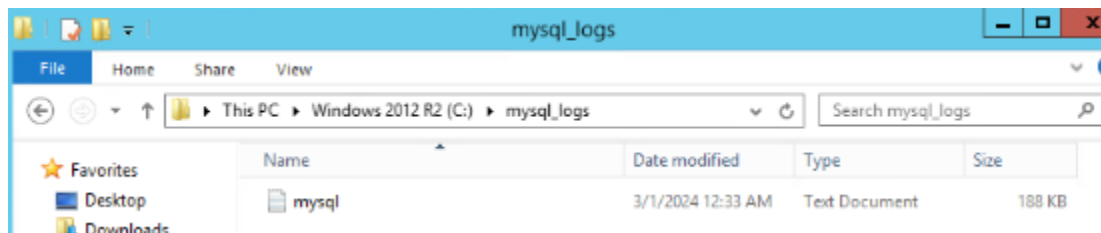
Notes:

- Most likely database that was compromised
- Is a verified sql backup on backup server
- Which system, service and users

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Approach, Steps and Actions Taken

First we logged into the database machine. Once there we opened file explorer and verified that the sql log was there –figure 1–.



When we opened it we realized that it was in a text file and quite difficult to read, meaning we would need to make it easier to read. For that we opened powershell and decided to use a `Select-String` to look for what had been done in the log file. We specifically looked for "Query" using `Select-String -Path C:\mysql_logs\mysql.log -Pattern "Query"` because we needed to find evidence of the attack. Query allowed us to look for actual SQL statements where interactions with that database occurred because a query is a request for information or data made by a user and written in a specific format. (Awati, 2024). What we found was several "Update" statements proving someone has been changing data specifically in critical Wordpress tables like the config table. –figure 2–

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```
PS C:\mysql_logs> Select-String -Path C:\mysql_logs\mysql.log -Pattern "Query"

mysql.log:5:2022-01-25T23:43:05.173960Z      2 Query      SET NAMES utf8mb4
mysql.log:6:2022-01-25T23:43:05.176818Z      2 Query      SET NAMES 'utf8mb4' COLLATE 'utf8mb4_unicode_520_ci'
mysql.log:7:2022-01-25T23:43:05.177529Z      2 Query      SELECT @@SESSION.sql_mode
mysql.log:8:2022-01-25T23:43:05.178710Z      2 Query      SET SESSION
mysql.log:10:2022-01-25T23:43:05.193580Z      2 Query      SELECT option_name, option_value FROM wp_options WHERE a
utoLoad =
'yes'
mysql.log:11:2022-01-25T23:43:05.590497Z      2 Query      SELECT option_value FROM wp_options WHERE option_name =
'siteorigin_panels_is_using_bundled' LIMIT 1
mysql.log:12:2022-01-25T23:43:05.689465Z      2 Query      SELECT option_value FROM wp_options WHERE option_name =
'wsmTimezoneString' LIMIT 1
mysql.log:13:2022-01-25T23:43:05.692363Z      2 Query      SHOW FULL COLUMNS FROM 'wp_options'
mysql.log:14:2022-01-25T23:43:05.695112Z      2 Query      UPDATE 'wp_options' SET 'option_value' = '1' WHERE 'opti
on_name'
= 'wsmKeepData'
```

Figure 2

Also by looking at the term user using `Select-String -Path C:\mysql_logs\mysql.log -Pattern "user"` we found large amounts of records indicating someone was deleting information from the `wp_usermeta`. This indicates that information may be missing from this log file and this may be the cause of the reported log in issues. Whoever was accessing the database was going through and deleting information from the `wp_usermeta` using sequential ID's which is not normal activity –figure 3.

```
mysql.log:1218:2022-01-25T23:47:10.536243Z      34 Query      SELECT user_id IN (4) ORDER BY umeta_id ASC
mysql.log:1219:2022-01-25T23:47:10.537584Z      34 Query      SELECT * FROM wp_usermeta WHERE umeta_id = 55
mysql.log:1220:2022-01-25T23:47:10.538973Z      34 Query      DELETE FROM 'wp_usermeta' WHERE 'umeta_id' = 55
mysql.log:1221:2022-01-25T23:47:10.575997Z      34 Query      SELECT * FROM wp_usermeta WHERE umeta_id = 56
mysql.log:1222:2022-01-25T23:47:10.582501Z      34 Query      DELETE FROM 'wp_usermeta' WHERE 'umeta_id' = 56
mysql.log:1223:2022-01-25T23:47:10.637604Z      34 Query      SELECT * FROM wp_usermeta WHERE umeta_id = 57
mysql.log:1224:2022-01-25T23:47:10.639134Z      34 Query      DELETE FROM 'wp_usermeta' WHERE 'umeta_id' = 57
mysql.log:1225:2022-01-25T23:47:10.701954Z      34 Query      SELECT * FROM wp_usermeta WHERE umeta_id = 58
mysql.log:1226:2022-01-25T23:47:10.703403Z      34 Query      DELETE FROM 'wp_usermeta' WHERE 'umeta_id' = 58
mysql.log:1227:2022-01-25T23:47:10.764277Z      34 Query      SELECT * FROM wp_usermeta WHERE umeta_id = 59
mysql.log:1228:2022-01-25T23:47:10.765848Z      34 Query      DELETE FROM 'wp_usermeta' WHERE 'umeta_id' = 59
mysql.log:1229:2022-01-25T23:47:10.826148Z      34 Query      SELECT * FROM wp_usermeta WHERE umeta_id = 60
mysql.log:1230:2022-01-25T23:47:10.827453Z      34 Query      DELETE FROM 'wp_usermeta' WHERE 'umeta_id' = 60
mysql.log:1231:2022-01-25T23:47:10.888639Z      34 Query      SELECT * FROM wp_usermeta WHERE umeta_id = 61
mysql.log:1232:2022-01-25T23:47:10.889920Z      34 Query      DELETE FROM 'wp_usermeta' WHERE 'umeta_id' = 61
```

Figure 3

We knew that based on the chat messages there was a backup file with verified integrity so we switched to the backup machine to examine it. In the backup machine and verified the sql logs existence by switching into the DatabaseBackup directory and listing the content there—Figure 4—.

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```
playerone@backup:~$ cd /DatabaseBackup
playerone@backup:/DatabaseBackup$ ls
wordpress.sql
```

Figure 4

We wanted to learn what users could have been potentially tampered with and because this was a safe state we could learn all of the users that would exist in the compromised system. We used grep to find the users in the file with `grep -n "wp_users" wordpress.sql`. “Wp_users” was used because that table “stores basic user information such as username, email, and password.” (User Insights, 2024) This showed us the 4 users in the file that were compromised, playerone, gbates, takasaka and admin— Figure 5—.

```
playerone@backup:/DatabaseBackup$ grep -n "wp_users" wordpress.sql
369:-- Table structure for table `wp_users`
372:DROP TABLE IF EXISTS `wp_users`;
375:CREATE TABLE `wp_users` (
394:-- Dumping data for table `wp_users`
397:LOCK TABLES `wp_users` WRITE;
398:/*140000 ALTER TABLE `wp_users` DISABLE KEYS */;
399:INSERT INTO `wp_users` VALUES (1,'playerone','$P$Bq1Ga4bFCJXKDK0kwHNg.dHwvVVHT9/', 'playerone', 'p
layerone@daswebs.com', '', '2019-11-04 06:58:22', '', 0, 'playerone'), (2, 'admin', '$P$B6Nb9KraeTnQ0Agw.JWP
ifhzbrygBZ/', 'admin', 'admin@daswebs.com', '', '2019-12-12 20:16:09', '', 0, 'Gary Thatcher'), (3, 'gbates',
'$P$B/BEQBtyakXwEbSPDqnnkqx7VXEn5/', 'gbates', 'gbates@daswebs.com', '', '2019-12-12 20:23:19', '', 0, 'Gi
lly Bates'), (4, 'takasaka', '$P$BYcJtX2cGPhn6QZPBtkj8mzWhCe.WX.', 'takasaka', 'takasaka@daswebs.com', '',
'2019-12-12 20:24:13', '', 0, 'Thanh Akasaka');
400:/*140000 ALTER TABLE `wp_users` ENABLE KEYS */;
playerone@backup:/DatabaseBackup$ _
```

Figure 5

We also found that you could find the same results plus additional account information by using `grep nickname ./wordpress.sql`. This information was found in the `wp_usermeta` table which hold additional user meta data in this case included things like the first and last name of the user associated with each account—figure 6— While in this case we only needed the user account names that were compromised, a more in depth investigation may require additional information which could be gathered in this manner.

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```
playerone@backup:/DatabaseBackup$ grep nickname ./wordpress.sql
INSERT INTO `wp_usermeta` VALUES (1,1,'nickname','playerone'),(2,1,'first_name',''),(3,1,'last_name',''),(4,1,'description',''),(5,1,'rich_editing','true'),(6,1,'syntax_highlighting','true'),(7,1,'comment_shortcuts','false'),(8,1,'admin_color','fresh'),(9,1,'use_ssl','0'),(10,1,'show_admin_bar_front','true'),(11,1,'locale',''),(12,1,'wp_capabilities','a:1:{s:13:\"administrator\";b:1;}'),(13,1,'wp_user_level','10'),(14,1,'dismissed_wp_pointers',''),(15,1,'show_welcome_panel','1'),(17,1,'wp_dashboard_quick_press_last_post_id','4'),(18,1,'community-events-location','a:1:{s:2:\"ip\";s:10:\"172.17.0.0\";}'),(19,1,'elementor_introduction','a:1:{s:10:\"rightClick\";b:1;}'),(20,1,'wp_user-settings','editor=html&siteorigin_panels_setting_tab=widgets&hidetb=1&libraryContent=browse'),(21,1,'wp_user-settings-time','1573154512'),(22,1,'so_panels_directory_enabled','1'),(23,2,'nickname','admin'),(24,2,'first_name','Gary'),(25,2,'last_name','Thatcher'),(26,2,'description',''),(27,2,'rich_editing','true'),(28,2,'syntax_highlighting','true'),(29,2,'comment_shortcuts','false'),(30,2,'admin_color','fresh'),(31,2,'use_ssl','0'),(32,2,'show_admin_bar_front','true'),(33,2,'locale',''),(34,2,'wp_capabilities','a:1:{s:13:\"administrator\";b:1;}'),(35,2,'wp_user_level','10'),(36,2,'dismissed_wp_pointers',''),(38,2,'session_tokens','a:1:{s:64:\"e038b8a78f81339b2283117ddd00149877595ba78196982938c19cd46e69c263\";a:4:{s:10:\"expiration\";i:1576354881;s:2:\"ip\";s:10:\"172.17.0.1\";s:2:\"ua\";s:78:\"Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:71.0) Gecko/20100101 Firefox/71.0\";s:5:\"login\";i:1576182081;}'}'),(39,2,'wp_dashboard_quick_press_last_post_id','104'),(40,2,'community-events-location','a:1:{s:2:\"ip\";s:10:\"172.17.0.0\";}'),(41,3,'nickname','gbates'),(42,3,'first_name','Gilly'),(43,3,'last_name','Bates'),(44,3,'description',''),(45,3,'rich_editing','true'),(46,3,'syntax_highlighting','true'),(47,3,'comment_shortcuts','false'),(48,3,'admin_color','fresh'),(49,3,'use_ssl','0'),(50,3,'show_admin_bar_front','true'),(51,3,'locale',''),(52,3,'wp_capabilities','a:1:{s:13:\"administrator\";b:1;}'),(53,3,'wp_user_level','10'),(54,3,'dismissed_wp_pointers',''),(55,4,'nickname','takasaka'),(56,4,'first_name','Thanh'),(57,4,'last_name','Akasaka'),(58,4,'description',''),(59,4,'rich_editing','true'),(60,4,'syntax_highlighting','true'),(61,4,'comment_shortcuts','false'),(62,4,'admin_color','fresh'),(63,4,'use_ssl','0'),(64,4,'show_admin_bar_front','true'),(65,4,'locale',''),(66,4,'wp_capabilities','a:1:{s:13:\"administrator\";b:1;}'),(67,4,'wp_user_level','10'),(68,4,'dismissed_wp_pointers','');
```

Figure 6

At this point we had all of the information we needed to conclude our investigation. The Database system was indeed breached, it was the mysql service, the log file with the evidence was mysql.log and playerone, gbates, takasaka and admin were the tampered with accounts. We input this information into the online form as requested and our investigation was complete.

Conclusion and Recommended Actions

Based on the evidence gathered during the investigation, we concluded that the database machine was the compromised system and that the MySQL service was the source of the incident. Review of the mysql.log file showed unauthorized SQL activity, including multiple UPDATE and DELETE statements against important WordPress tables. These actions affected both site configuration data and user account information, which explains the login issues reported in the scenario. Additional review of the known-good backup confirmed that several user accounts had

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been tampered with, specifically playerone, gbates, takasaka, and admin. Taken together, the evidence supports the conclusion that the WordPress environment experienced a database-level compromise that impacted authentication and account integrity.

The most practical response would be to restore the database from the verified backup, since it provides a trusted version of the affected records and configuration data. After restoration, all impacted and privileged accounts should have their passwords reset, and access to the database should be reviewed to ensure only authorized users and services have permission. In addition, the organization should preserve and review related database, web, and system logs to determine how the attacker originally gained access and whether any additional systems were affected. To reduce the likelihood of this happening again, administrators should apply security updates, strengthen access controls, enable stronger monitoring for suspicious SQL activity, and continue maintaining validated backups for rapid recovery. These actions would both remediate the current incident and improve the organization's ability to detect and respond to future compromises.

Three and two debrief

For this lab overall it went fairly well, it didn't take too long you can look at the exported PDF and see that the actual flag was completed a while before the lab was turned in this was due to writing and that's one thing that we could do better in the future is right as we go instead of waiting till the end. He also got interrupted on both ends so that added some additional time. One other thing that we could do better of in the future is making sure we get all the required screenshots, We realized after submitting it that we forgot to take several screenshots so I reopened one and took a couple of needed screenshots only to realize after our scheduled

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window had closed that we also forgot to take a screenshot of the answers that we had submitted into the online form. It's clear that we submitted the online form based on having a completed PDF with all the flags checked but we don't have a screenshot of it in the lab.

Now on to three things that went well. One thing that went well is that we had talked to other people in the class who warned us ahead of time to make sure we were checking our syntax as that was something that had slowed them down so we were more conscious of our syntax going into the lab. They didn't tell us how to do the lab or what the answers were, having additional tips for making sure we were checking the syntax for commands was very helpful. This was in line with the environmental chat log saying that it was likely the database that was compromised because that gave a more clear area of where to focus our attention.

The second thing that went well in this lab was that it didn't take nearly as long as we expected, we had planned for it to take much longer when it came to finding a time to do it so it gave us a bit of relief to have it done earlier than anticipated. We also had different thought processes when it came to what to look for, for example one partner was operating under the assumption that it was the SQL database that was tampered with which ended up being correct while the other partner made sure to look at some of the contents of the file to confirm that it was likely the one that was compromised So even though this took a little bit more time it Was verifying our thought process. The difference in thought process could also be viewed when we were using the grub command. One of us looked for a nickname while one of us looked for the WordPress user table, both of which had the correct information. It was just different ways of getting there so having different thought processes worked well in this lab.

The third thing that went well in this lab was fairly straight forward, this is somewhat connected with that the lab didn't take us very long but in other labs we've had issues where we

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think we found the right answer but it turns out we didn't and we have to submit the form multiple times but in this one each answer we found was correct on the 1st attempt. Going into the lab we still looked for all possible answers even though we already knew that it was likely the database system that was compromised we still verified it Same with how we figured it was going to be the SQL log that showed the evidence but we still needed to verify it We went in having fairly good idea of what the answers were going to end up being but it was simple to verify them and it was not a very complicated process which helped a lot in alleviating some of the stress as we were worried going into the lab that it would be a lot more difficult.

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NICE Challenge Project

Challenge Submission Report

<https://portal.nice-challenge.com/reports/verify/07E60-FFF1-A52A4/>

Submission ID: 154126

Timestamp: 3/29/2026 9:22 PM UTC

Name: Anna Fanco

Challenge ID: 168

Challenge Title: Lengthy Logs: Attack Analysis [NG]



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Scenario

Lately, employees have been having some issues logging in to one of our Wordpress websites and we can't figure out why. Our security analyst suspects that we might have been hit by a cyber attack, but is currently indisposed and can't look into it. I need you to take a look and confirm whether or not we were actually hit and if so, what the impact of that attack might have been.

Duration

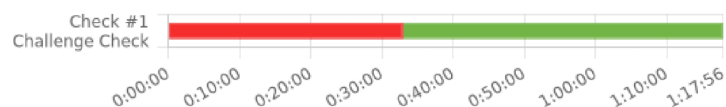
1:17

Final Check Details

✓ Check #1: Incident Response Form Correctly Submitted

Full Check Pass

Full: 1/1



Element

Cybersecurity

Work Role

Cyber Defense Analyst

DCWF Tasks

• 750: Perform event correlation using information gathered from a variety of sources within the enterprise to gain situational awareness and determine the effectiveness of an observed attack.

Knowledge, Skills, and Abilities

- 22: Knowledge of computer networking concepts and protocols, and network security methodologies.
- 34: Knowledge of database systems.
- 61: Knowledge of incident response and handling methodologies.
- 63: Knowledge of cybersecurity principles and organizational requirements (relevant to confidentiality, integrity, availability, authentication, non-repudiation).
- 66: Knowledge of intrusion detection methodologies and techniques for detecting host and network-based intrusions via intrusion detection technologies.
- 70: Knowledge of information technology (IT) security principles and methods (e.g., firewalls, demilitarized zones, encryption).
- 81A: Knowledge of network protocols such as TCP/IP, Dynamic Host Configuration, Domain Name System (DNS), and directory services.
- 90: Knowledge of operating systems.
- 92: Knowledge of how traffic flows across the network (e.g., Transmission Control Protocol [TCP] and Internet Protocol [IP], Open System Interconnection Model [OSI], Information Technology Infrastructure Library, current version [ITIL]).
- 105: Knowledge of system and application security threats and vulnerabilities (e.g., buffer overflow, mobile code, cross-site scripting, Procedural Language/Structured Query Language [PL/SQL] and injections, race conditions, covert channel, replay, return-oriented attacks, malicious code).
- 139: Knowledge of the common networking protocols (e.g., TCP/IP), services (e.g., web, mail, Domain Name Server), and how they interact to provide network communications.
- 150: Knowledge of what constitutes a network attack and the relationship to both threats and vulnerabilities.
- 278: Knowledge of different types of network communication (e.g., LAN, WAN, MAN, WLAN, WWAN).
- 342A: Knowledge of operating system command line/prompt.
- 353: Skill in collecting data from a variety of cyber defense resources.
- 895: Skill in recognizing and categorizing types of vulnerabilities and associated attacks.
- 1121: Knowledge of Windows/Unix ports and services.
- 1158: Knowledge of cybersecurity principles.
- 1159: Knowledge of cyber threats and vulnerabilities.

Centers of Academic Excellence Knowledge Units

- Cybersecurity Foundations
- Cybersecurity Principles
- Cyber Threats
- Operating Systems Administration
- Operating Systems Concepts
- Vulnerability Analysis

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References

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